

HAR - Facial Expression

Sofía Jimeno Lucía
International Master in Computer Vision
Universidade do Porto
Porto, Portugal
up202200682@edu.fe.up.pt

I. INTRODUCTION

This project aims to tackle the interesting task of emotion recognition from facial expressions, a critical domain within the field of computer vision and affective computing. To achieve this goal, we turn to the Facial Expression Recognition dataset FER-2013, which is a collection of 35,887 images categorised into seven different emotional states: from anger and disgust to fear, happiness, sadness, surprise and neutrality.

However, it's crucial to recognise the inherent class imbalance within the dataset, which poses a significant challenge to the development of accurate emotion recognition models. In this project, we address this imbalance using advanced classification techniques, leveraging the ResNet50V2 architecture to overcome the complexity of image classification and improve model performance.

II. DATASET

The dataset proposed for this project is Facial Expression Recognition dataset FER-2013. The data consists of 48x48 pixel grayscale images of faces. The faces have been automatically registered so that the face is more or less centred and occupies about the same amount of space in each image.

The Figure 1 shows samples of images of the dataset corresponding to the different categories: 0=Angry, 1=Disgust, 2=Fear, 3=Happy, 4=Sad, 5=Surprise, 6=Neutral. The training set consists of 28.710 examples and the public test set consists of 7.178 examples.

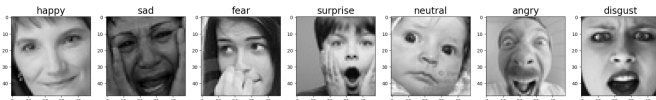


Fig. 1. Examples of the images for each category

The Figure 2 shows the distributions of the classes for both the training set and the test set. There is a clear imbalance with two of the categories: "Disgust" (far below the rest) being only the 1.5% of the images and "Happy" (way ahead of the rest) with a 25% approximately.

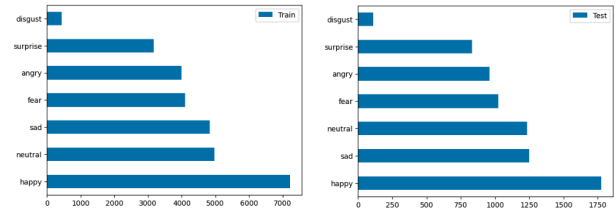


Fig. 2. Classes distribution

III. METHODOLOGY

A. Data preparation

Data augmentation was applied in the images to increase the diversity and quantity of training data, in order to improve the generalization and robustness of the model by exposing them to a wider range of variations and scenarios.

To carry out the data augmentation, the method *ImageDataGenerator* [1] for applying random transformations was used, specifically the selected ones:

- rotation range (equal 10). Randomly rotate the image between 0 and 10 degrees.
- zoom range (equal 0.2). This method randomly zooms the image either by zooming in or it adds some pixels around the image to enlarge the image.
- width shift range (equal 0.1). It shifts randomly the image to the left or right (horizontal shifts).
- height shift range (equal 0.1). It works same as width shift but shift vertically (up or down).
- horizontal flip (equal True). It means reversing the rows or columns of pixels in the case of a vertical or horizontal flip respectively.
- fill mode (equal nearest). When your image shift, there is some space left over. You can select a few types of methods to fill the area:
 - constant — Which will fill the area in black color. You can change the color by giving a value for cval
 - nearest — Which will fill the area with the nearest pixel and stretching it.
 - reflect — Which will fill the area with the reflection of the image.
 - wrap — Which will fill the area by wrapping the image around it.

In addition to applying transformations, the original training dataset was divided into two subsets to facilitate the creation of validation images, where said subset comprising 25% of the total dataset.

B. Classifier: ResNet-50 v2

ResNet50V2 belongs to the family of ResNet models, which are renowned for their effectiveness in training very deep neural networks without encountering the vanishing gradient problem.

The ResNet architecture follows two basic design rules. First, the number of filters in each layer is the same depending on the size of the output feature map. Second, if the feature map's size is halved, it has double the number of filters to maintain the time complexity of each layer.

1) Residual Learning

ResNet50V2 employs residual learning, which involves introducing shortcut connections, also known as skip connections, that skip one or more layers. These connections facilitate the flow of gradients during training, allowing for smoother optimization. Residual blocks, the fundamental building blocks of ResNet architectures, consist of these skip connections along with convolutional layers.

2) Bottleneck Design:

The architecture of ResNet50V2 incorporates bottleneck blocks, which are composed of three layers: 1x1, 3x3, and 1x1 convolutions. This design helps in reducing the computational complexity and memory footprint of the model while preserving representational capacity. The 1x1 convolutions are responsible for reducing and then increasing the dimensions, while the 3x3 convolution layer captures spatial features.

3) Version 2 Enhancements:

ResNet50V2 introduces several enhancements over its predecessor, ResNet50. These enhancements include using pre-activation units within each residual block, which involves applying batch normalization and ReLU activation before convolution operations. This adjustment mitigates the degradation problem observed in very deep networks and improves training convergence.

4) Global Average Pooling and Softmax

Towards the end of the ResNet50V2 architecture, after the convolutional layers, global average pooling is applied to reduce the spatial dimensions of the feature maps into a single vector. This vector is then fed into a fully connected layer followed by a softmax activation function, which generates probabilities for each class in the classification task.

```
{0: 1.0264550264550265,
1: 9.407601572739187,
2: 1.0010692203988656,
3: 0.5685244343533015,
4: 0.8260702777351542,
5: 0.8490990102913923,
6: 1.2931003422806702}
```

Fig. 3. Class weights

IV. EXPERIMENTS

This section will explain the various experiments which have been carried out in an attempt to get better results in the classification.

A. Data augmentation - Maximize accuracy

The initial experiment was done by using the data augmentation described in the methodology section and using as metric the accuracy for maximizing the Resnet50.

B. Data augmentation - Maximize F1 score

For the second experiment, I decided to use F1 score instead of accuracy because of the class imbalance scenario.

Accuracy can be misleading when classes are unbalanced. For example, if a dataset has 95% of samples from one class and only 5% from another, a model that predicts the majority class for all samples would still achieve 95% accuracy. However, it will not capture the performance on the minority class. The F1 score, which takes into account both precision and recall, provides a more balanced assessment of the model's performance across all classes.

C. Data augmentation - Class weights

Again, to address issues of class imbalance in the training data, I decided to try class weights, shown in Figure 3, to help improve the model's performance.

If certain classes are underrepresented compared to others, the neural network may be biased towards the majority class during training, leading to poor performance on the minority classes. By assigning higher weights to the minority classes during training, the neural network learns to pay more attention to these classes, resulting in a more balanced and accurate model.

D. Without data augmentation

In this experiment, the images didn't undergo any transformation.

E. Over-sampling and under-sampling

Another attempt to overcome the problem of an imbalanced dataset and to avoid bias towards the majority class was to use over- and under-sampling techniques.

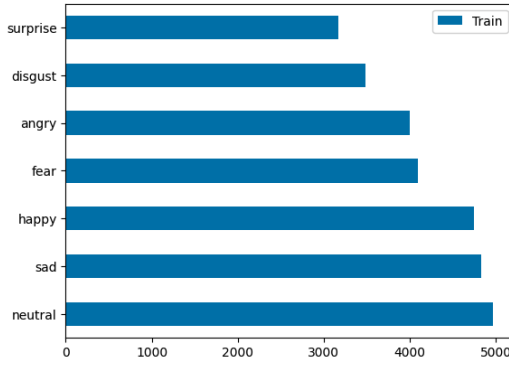


Fig. 4. Balanced distribution

Looking again at the initial distribution, we can see that the imbalanced classes are "Disgust" and "Happy". To correct for these differences, I decided to use over-sampling for "Disgust" and under-sampling for "Happy". For the first case, the images of this class were duplicated 8 times and different transformations were randomly applied: flip, rotate, noise or/and blur. For the second case, only 4747 images were randomly selected from the original folder. In Figure 4, the final distribution of the balanced samples are shown.

V. RESULTS

Table I shows the results for the different experiments. Overall, the best model was the first one, where the data augmentation was applied over the images and the accuracy was used for maximisation.

For experiments 2, 3, 4 and 5, the model started to overfit in early epochs, meaning that the metrics in training, such as loss and accuracy, improved but not in the validation set.

On the other hand, in the first experiment the model was very stable during the 20 epochs it was trained. Finally, the last experiment didn't show any overfitting, but it also didn't improve on the first one, achieving an accuracy of 67%.

TABLE I
RESULTS OF THE FACIAL EXPRESSION CLASSIFICATION EXPERIMENTS

ID	Description	Loss	Accuracy	F1
1	Data augmentation with max. accuracy	0.9179	0.6818	0.441
2	Data augmentation with max. f1	1.2545	0.5582	
3	Data augmentation with weights			
4	Non data augmentation	1.1358	0.6532	
5	Non data augment. + Over-sampling and under-sampling	1.2244	0.6510	
6	Data augmentation + Over-sampling and under-sampling	0.9492	0.6730	

I decided to include only the plot of the best of the models (Experiment 1) for not adding noise to the report. In Figure 5, the learning curves in "Train" and "Val", for Loss and

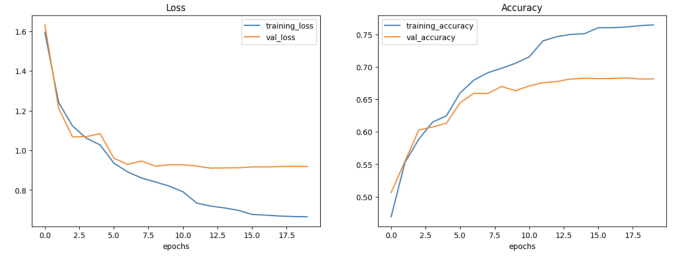


Fig. 5. Loss and Accuracy of Experiment 1 (best of the models)

Accuracy, of the model are shown.

In Figure 6, the metrics for each of the classes are shown, like so a macro and a weighted average of all of them. The precision scores vary across classes, with the highest precision achieved for the "Happy" class (0.88) and the lowest for "Fear" (0.54). Similarly, there are also differences for the recall scores, with "Happy" having the highest recall (0.88) and "Disgust" having the lowest (0.48). The F1-scores, which balance precision and recall, show the highest performance for "Happy" (0.88) and the lowest for "Disgust" (0.55).

	precision	recall	f1-score	support
Angry	0.61	0.64	0.62	958
Disgust	0.66	0.48	0.55	111
Fear	0.54	0.46	0.49	1024
Happy	0.88	0.88	0.88	1774
Neutral	0.62	0.69	0.66	1233
Sad	0.55	0.56	0.56	1247
Surprise	0.80	0.77	0.79	831
accuracy			0.68	7178
macro avg	0.67	0.64	0.65	7178
weighted avg	0.68	0.68	0.68	7178

Fig. 6. Metrics for class of Experiment 1 (best of the models)

In Figure 7, the Confusion Matrix is shown. The diagonal elements indicate correct predictions, while off-diagonal elements represent misclassifications. Analyzing the matrix reveals that the model performs well for the "Happy" class, with a high number of true positives (1566) and relatively low false positives and false negatives. However, it struggles with "Neutral" and "Sad" emotions, as evidenced by the misclassifications into other classes, particularly "Fear" and "Angry". The "Disgust" class also suffers from misclassifications, with only 53 true positives and notable confusion with "Angry" and "Sad".

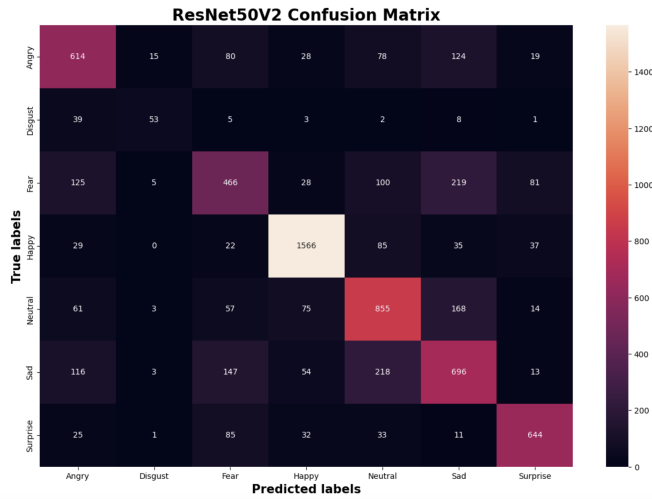


Fig. 7. Confusion Matrix of Experiment 1 (best of the models)

Some examples of facial recognition of our model are shown in Figures 8 and 9.

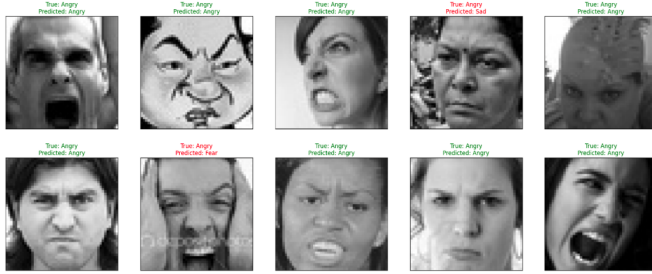


Fig. 8. Pictures from the dataset with their labels (real and predicted)

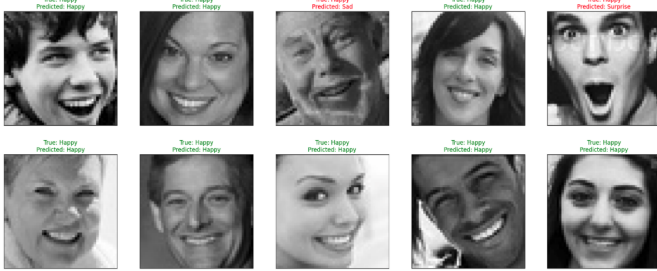


Fig. 9. Pictures from the dataset with their labels (real and predicted)

VI. DISCUSSION AND CONCLUSIONS

As it has been already mention before, the main challenge of training a good facial recognition classifier in this case is the imbalanced data.

After trying different techniques and doing several experiments, the best of the model gotten has a 68% of Accuracy. The model shows strong performance for "Happy" and "Surprise" classes, it struggles with "Disgust" and "Fear," indicating potential areas for improvement in class prediction.

Some possible solutions to mitigate class imbalances and improve the model's ability to discern between different emotions might be: augmenting the dataset with more additional real example; enhancing the resolution of the images, providing finer details of facial expressions, aiding in better feature extraction and classification; or even clarifying ambiguous categories within the dataset can contribute to more precise emotion labeling and reduce misclassifications.

REFERENCES

- [1] ImageDataGenerator: <https://www.tensorflow.org/tf/keras/preprocessing/image/ImageDataGenerator> Documentation:
- [2] ResNet50V2: <https://www.tensorflow.org/tf/keras/applications/ResNet50V2> Documentation: